WORKED SOLUTION

STEP II 1995, Question 12

Probability and Statistics • Sequential Duels

STEP II 1995, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.13, *The Truel*: https://beyondstep.keithmcnulty.org/ch12/truel.html

The key idea

In a duel that goes on until someone is hit, the game "starts again" every time both players miss. So the probability we want satisfies a simple equation in terms of itself, and there's no need to sum an infinite series (though you can). For the three-way duel, each option for Algernon's first shot quickly turns into a two-person duel, and the first result handles those. The surprise is that Algernon's best move is to hit nobody.

Solution

The two-person duel

Let P be the probability that Percy wins when he throws first. Look at the first pair of throws:

- Percy hits with his first throw (probability p), and he wins at once;
- Percy misses and then Rodney hits (probability $(1-p)r$), and Percy loses;
- both miss (probability $(1-p)(1-r)$), and we are back where we started with Percy to throw. Throws are independent, so from here Percy wins with probability P again.

Hence

$$P = p + (1-p)(1-r)P \quad\Longrightarrow\quad P\big(1-(1-p)(1-r)\big) = p \quad\Longrightarrow\quad P = \frac{p}{p+r-pr},$$

since $1-(1-p)(1-r) = p+r-pr$, which is positive because $p > 0$. This is the required result.

Ask yourself Does the game definitely end, so that Percy losing really is the same as Rodney winning?

Yes. The chance that the first n rounds all end in double misses is $\big((1-p)(1-r)\big)^n$, which tends to 0 because $p > 0$. Equivalently, summing the geometric series $\sum_{k\geq 0} p\big((1-p)(1-r)\big)^k$ gives the same answer.

In the three-way duel, A, B and C hit with probabilities $\frac{2}{5}$, $\frac{3}{5}$ and 1. A two-person duel between A and B with A throwing first will come up twice, so we work it out now using the result above with $p = \frac{2}{5}$, $r = \frac{3}{5}$:

$$\frac{\frac{2}{5}}{\frac{2}{5}+\frac{3}{5}-\frac{6}{25}} = \frac{\frac{2}{5}}{\frac{19}{25}} = \frac{10}{19}.$$

Part (i)

If Algernon hits Cuthbert, then only Algernon and Bertie are left, and Bertie throws next, aiming at Algernon. Algernon survives only if Bertie misses (probability $\frac{2}{5}$). After that it is a duel between A and B with A throwing first, which Algernon wins with probability $\frac{10}{19}$. So

$$\mathrm{P}(\text{A wins}) = \frac{2}{5}\cdot\frac{10}{19} = \frac{4}{19}.$$

Part (ii)

If Algernon hits Bertie, Cuthbert throws next with only Algernon left to aim at. Cuthbert never misses, so Algernon's chance of winning is 0.

Part (iii)

If Algernon misses, all three are still in. Bertie throws next, and aims at Cuthbert.

- With probability $\frac{3}{5}$ Bertie hits Cuthbert. Algernon and Bertie are left with Algernon to throw, and Algernon wins with probability $\frac{10}{19}$.
- With probability $\frac{2}{5}$ Bertie misses. Cuthbert then aims at Bertie and is certain to hit him. Algernon and Cuthbert are left with Algernon to throw. Algernon must hit with this throw, because otherwise Cuthbert hits him. So he wins with probability $\frac{2}{5}$.

Therefore

$$\mathrm{P}(\text{A wins}) = \frac{3}{5}\cdot\frac{10}{19} + \frac{2}{5}\cdot\frac{2}{5} = \frac{6}{19} + \frac{4}{25} = \frac{150+76}{475} = \frac{226}{475}.$$

This is the same however the miss comes about, whether deliberate or a failed aim, because Bertie and Cuthbert play the same way afterwards.

Algernon's best plan

Ask yourself Algernon can't choose to *hit* someone, only to *aim* at them. What does each choice really give him?

If he aims at someone, he hits with probability $\frac{2}{5}$ and misses with probability $\frac{3}{5}$, and a miss leads to part (iii). So:

$$\begin{aligned}
\text{aim at Cuthbert:} &\quad \frac{2}{5}\cdot\frac{4}{19}+\frac{3}{5}\cdot\frac{226}{475}=\frac{200+678}{2375}=\frac{878}{2375}\approx 0.370,\\
\text{aim at Bertie:} &\quad \frac{2}{5}\cdot 0+\frac{3}{5}\cdot\frac{226}{475}=\frac{678}{2375}\approx 0.285,\\
\text{deliberately miss:} &\quad \frac{226}{475}=\frac{1130}{2375}\approx 0.476.
\end{aligned}$$

Even a certain hit on Cuthbert would only give $\frac{4}{19}=\frac{100}{475}$, which is less than $\frac{226}{475}$. Hitting either opponent can only lower his chances. Algernon should therefore **deliberately miss with his first shot**. Then Bertie and Cuthbert shoot at each other, and Algernon gets the first throw against whoever survives. With this plan his probability of winning is $\frac{226}{475}$, as required.

Checks

- If $r=0$ the duel formula gives 1, and if $p=1$ it gives 1, both as they should.
- The three options rank as $0.476 > 0.370 > 0.285$. A Monte Carlo simulation of the whole three-way duel (3×10^5 games per strategy) gave 0.476, 0.369 and 0.284, matching the exact values.
- Part (iii) as a sanity check: $\frac{6}{19}\approx 0.316$ and $\frac{4}{25}=0.16$ add to 0.476.

Common slips

- Getting the order of throws wrong after the first shot. In part (i) Bertie throws *next*, so the duel formula only applies after Bertie has missed.
- Treating "aim at Cuthbert" as if it meant "hit Cuthbert". Parts (i)–(iii) are conditional on the outcome, and the plan has to weight them by $\frac{2}{5}$ and $\frac{3}{5}$.

Beyond STEP

Writing a probability in terms of itself after one step is called *first-step analysis*. It is the basic tool for Markov chains, where it gives hitting probabilities and expected hitting times as solutions of linear equations. The "truel" is a well-known puzzle in game theory, and the weakest player often does best by staying out of the fight. For more, see

Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*.

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